



## MATHEMATICS FORM ONE MARKING SCHEME

1.40000      1mk

2. a)  $30816 \div 24 = 1284$  cartons      1mk

b)  $1284 \times 2 = 2568$

c) Mass of 24 books  $= (12 \times 2) = 10$  kg

$$\frac{30816 \times 10}{24} \text{ m1}$$

$$= 12840 \text{ kg A1}$$

3. Numerator

$$= 8 \div 2 \times 9 - 4 \times 6$$

$$= 4 \times 9 - 4 \times 6$$

$$= 36 - 24$$

$$= 12 \quad \text{M1}$$

Denominator

$$= 56 \div 7 \times 2$$

$$= 8 \times 2$$

$$= 16 \quad \text{M1}$$

$$= 12/16$$

$$= \frac{3}{4} \quad \text{A1}$$



$$4. 113400 = 2 \times 56700$$

$$= 2 \times 2 \times 28350$$

$$= 2 \times 2 \times 2 \times 14175$$

$$= 2 \times 2 \times 2 \times 3 \times 4725$$

$$= 2 \times 2 \times 2 \times 3 \times 3 \times 1575 \quad \text{M1}$$

$$= 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 525$$

$$= 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 175$$

$$= 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 5 \times 5 \times 7$$

$$= 2^2 \times 3^4 \times 5^2 \times 7 \quad \text{A1}$$

$$5. \quad 0.36 \times 1000 = 360g$$

$$0.504 \times 1000 = 504g \checkmark$$

$$0.672 \times 1000 = 672g$$

|   |     |     |     |
|---|-----|-----|-----|
| 2 | 360 | 504 | 672 |
| 2 | 180 | 252 | 336 |
| 2 | 90  | 126 | 168 |
| 3 | 45  | 63  | 84  |
|   | 15  | 21  | 28  |

M1

$$= 24g$$

$$= \frac{24}{1000} = 0.024kg \quad \text{A1}$$

$$6. \quad 181 - 5 = 176$$

$$236 - 5 = 231 \quad \text{1mk}$$



|    |     |     |
|----|-----|-----|
| 11 | 176 | 231 |
|    | 16  | 21  |

$$=11\sqrt{\quad}$$

$$7. \text{Number} = \frac{G.C.D \times L.C.M}{\text{Given Number}} \sqrt{\quad}$$

$$a=60$$

$$b=150$$

$$c = \frac{30 \times 900}{60} = 450 \sqrt{\quad}$$

$$d = \frac{30 \times 900}{150} = 180 \sqrt{\quad}$$

$$8.a) (-7) + (-2) + (+6)$$

$$=-3$$

$$b) (-9) - (-12)$$

$$=-21$$

9. Man born 1966

Father born 1928

Mother born  $1928+3=1931 \sqrt{\quad}$

Daughter born 1992

Son  $1992-5=1987 \sqrt{\quad}$

$$1987-1931=56 \sqrt{\quad}$$

10. 97410

$$\frac{410}{8} = 51 \text{rem} 2 \sqrt{\quad}$$



97410 is not divisible by 8 since the last three digits are not divisible by 8.  $\checkmark$